

HemoScan AI - Clinical Hematology Report

Clinical Hematology Report

Patient Information:

- * Name: Piyush Kumar Singh
- * Age: 21.0 Years
- * Gender: Male
- * Phone: 9876543210
- * Report Status: ML-Assisted Analysis (Confidence: 96.83%)

Complete Blood Count (CBC) Comparison Table

Parameter	Patient Value	Reference Range (Adult Male)	Status
Hemoglobin (Hb)	11.5 g/dL	13.5 – 17.5 g/dL	Low
RBC Count	4.1 x 10 ¹² /μL	4.5 – 5.9 x 10 ¹² /μL	Low
PCV (Hematocrit)	34.2 %	41.0 – 50.0 %	Low
MCV	82.0 fL	80.0 – 100.0 fL	Normal (Borderline)
MCH	26.5 pg	27.0 – 33.0 pg	Low
MCHC	31.5 g/dL	32.0 – 36.0 g/dL	Low

Clinical Interpretation

The laboratory findings for Mr. Piyush Kumar Singh indicate a state of Anemia, as evidenced by the Hemoglobin level of 11.5 g/dL, which falls significantly below the standard physiological threshold for an adult male (13.5 g/dL). The Red Blood Cell (RBC) count and Packed Cell Volume (PCV) are also reduced, confirming a decrease in the oxygen-carrying capacity of the blood.

A detailed analysis of the erythrocyte indices reveals a Hypochromic state. While the MCV (Mean Corpuscular Volume) is 82.0 fL—placing it at the very low end of the normal range—the MCH (Mean Corpuscular Hemoglobin) and MCHC (Mean Corpuscular Hemoglobin Concentration) are both below normal limits. This suggests that although the red blood cells are currently normal in size (normocytic), they are deficient in hemoglobin content (hypochromic). This pattern is frequently observed in the early stages of nutritional deficiencies or chronic conditions before the cell size significantly diminishes.

Risk Level: Moderate

The patient's condition is classified as Moderate Risk. While a hemoglobin level of 11.5 g/dL is not immediately life-threatening (which typically occurs below 7.0–8.0 g/dL), it is clinically significant for a 21-year-old male. If left unaddressed, it may lead to symptoms such as chronic fatigue, dyspnea (shortness of breath) on exertion, tachycardia, and reduced cognitive concentration. The high confidence score of the ML model (96.83%) underscores the necessity for a formal diagnostic workup.

Possible Anemia Type

Based on the borderline MCV and low MCH/MCHC, the differential diagnoses include:

- Early-Stage Iron Deficiency Anemia (IDA): This is the most common cause where hemoglobin synthesis is impaired before the cells become overtly microcytic (small).
- Anemia of Chronic Disease (ACD): Often presents as normocytic/normochromic but can transition into a hypochromic pattern.

3. Sideroblastic Anemia: Less common, but involves a failure to incorporate iron into hemoglobin.

Suggested Diagnostic Tests

To pinpoint the etiology of the anemia and initiate appropriate therapy, the following investigations are recommended:

1. Iron Profile: Including Serum Iron, Total Iron Binding Capacity (TIBC), and Serum Ferritin (to evaluate iron stores).
2. Peripheral Blood Smear (PBS) Examination: To visualize red cell morphology (checking for anisocytosis or poikilocytosis).
3. Reticulocyte Count: To assess the bone marrow's regenerative response to the anemia.
4. Vitamin B12 and Folate Levels: To rule out concurrent nutritional deficiencies that might be masking a more severe microcytosis.
5. Stool Routine/Occult Blood: To rule out any gastrointestinal blood loss, which is a common cause of iron depletion in young adults.